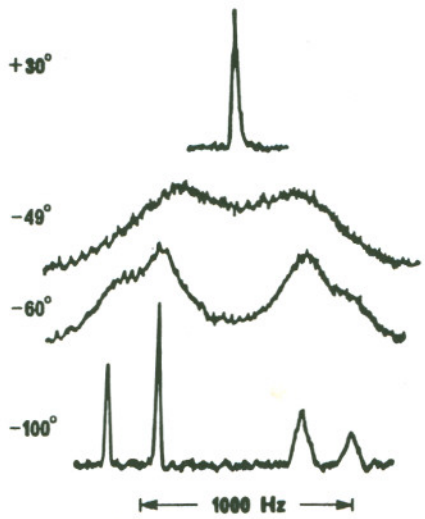
PHYSICAL CHEMISTRY CUMULATIVE EXAM
February 9, 2001

(1) The F-19 NMR spectra of 1,1-difluorocyclohexane was obtained as a function of temperature (see below). Explain the temperature effect and the origin of the four lines observed at –100°.



(2) Draw the complete H-1 NMR spectrum of $Si(CH_3)_4$ obtained on a 500 MHz spectrometer. Draw the complete C-13 spectrum of this molecule with and without H-1 decoupling. Draw the complete Si-29 spectrum of this molecule with and without H-1 decoupling.

H-1 [Resonance Frequency = 500 MHz, Spin = ½, Natural Abundance = 99.985%]

C-13 [Resonance Frequency = 125.759 MHz, Spin = ½, Natural Abundance = 1.108%]

Si-29 [Resonance Frequency = 99.345 MHz, Spin = -½, Natural Abundance = 4.7%]